

A learning analytics-based leaderboard feedback approach for promoting student cognitive engagement and learning performance in online collaborative learning

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Cognitive engagement is crucial for achieving positive learning outcomes. However, it is often inadequate in online collaborative learning. While learning analytics feedback can promote learners' engagement, it may have limitations in motivating students to continue participating. As a gamification element, the leaderboard has been shown to boost learning motivation, but its effects in conjunction with learning analytics feedback have not been extensively investigated. This study proposed a learning analytics-based leaderboard feedback approach (LALF) and conducted a quasi-experimental study involving 32 engineering students to assess the impact of this approach on student cognitive engagement and their learning performance. The experimental group received LALF, while the control group only received the learning analytics feedback. Utilizing chi-squared tests, epistemic network analysis (ENA) and auto-recurrence quantification analysis (aRQA), we examined the effects of LALF on the distributions, patterns, and dynamics of cognitive engagement. The results indicated that students in the experimental group exhibited significantly higher high-level cognitive engagement behaviours than those in the control group. Furthermore, students in the experimental group who engaged with the LALF tended to exhibit stronger connections among high-level cognitive engagement

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behaviours and more stable cognitive engagement patterns than those in the control group. Additionally, the results showed that students in the experimental group achieved higher learning performance than those in the control group. These findings reveal the critical role of combining learning analytics feedback with leaderboards in enhancing cognitive engagement in online collaborative learning, providing important guidance for designing efficient online learning experiences and improving educational quality.

KEYWORDS

aRQA, cognitive engagement, ENA, leaderboard, learning analytics feedback, online collaborative learning

INTRODUCTION

Online Collaborative Learning (OCL) has emerged as a key pedagogical approach in modern education, enabling learners to co-construct knowledge through continuous interactions. Research demonstrates that OCL can enhance students' critical thinking and facilitate deep learning (Koşar, 2021; Magen-Nagar & Shonfeld, 2018). Cognitive engagement—students' active efforts in knowledge construction—serves as a key indicator of OCL effectiveness, directly enhancing learning outcomes (Farrow et al., 2021; Kim et al., 2020; Liu et al., 2022). However, maintaining high-level cognitive engagement in OCL remains challenging without targeted support (Baanqud et al., 2020; Camacho et al., 2020; Chen et al., 2018, 2022; Wu et al., 2025).

Recent research has emphasized learning analytics feedback as a means to enhance student engagement by providing insights into students' behaviours, cognitive processes and social interactions (Chen et al., 2018, 2022; Ouyang & Zhang, 2024). However, as noted in the literature, its effectiveness may depend on how students adopt and engage with it, and studies have suggested that additional motivational support or complementary strategies may be necessary (Chen & Chen, 2023; Jivet et al., 2017; Strauß & Rummel, 2021). In this context, leaderboards—a widely used gamification element—have been introduced as a supplementary strategy to address these motivational limitations of learning analytics feedback by fostering peer comparison and competition, thereby potentially enhancing students' cognitive engagement (Balci et al., 2022; Chen & Chen, 2023; Cigdem et al., 2024; Moon et al., 2024). However, so far, few studies have investigated the effects of integrating learning analytics feedback and leaderboards on cognitive engagement, particularly in OCL environments.

Therefore, this study proposes a learning analytics-based leaderboard feedback approach (LALF) that synergistically combines the data-driven capabilities of learning analytics feedback with gamified leaderboard elements. Using a quasi-experimental design, we investigate the effect of the LALF on students' cognitive engagement and learning performance in OCL through combined temporal analysis and traditional statistical methods.

The following section elaborates on the literature on the relevant topics, encompassing both theoretical background and extant research findings, leading to explicit research questions and hypotheses. Subsequently, the research methodology is delineated, followed by results, discussion and conclusion.

Practitioner notes

What is already known about this topic?

- Cognitive engagement is essential for achieving positive learning outcomes, particularly in online collaborative learning environments.
- Learning analytics feedback can enhance learner engagement, but may lack elements that stimulate motivation among students.
- The leaderboard is considered a gamification element, potentially boosting learning motivation by fostering a competitive atmosphere.

What this paper adds?

- This study introduces a learning analytics-based leaderboard feedback approach (LALF), which combines learning analytics feedback and leaderboards.
- It provides empirical evidence from a quasi-experimental design involving 32 engineering students, indicating that students who engaged with the LALF demonstrated higher levels of cognitive engagement behaviours compared to those who only received traditional learning analytics feedback.
- The study employs various analytical methods, including chi-squared tests, epistemic network analysis (ENA) and automated recurrence quantification analysis (aRQA), to explore the distributions, patterns and dynamics of cognitive engagement associated with the LALF.

Implications for practice and policy

- Educators may want to consider integrating leaderboards with learning analytics feedback to foster a competitive yet supportive online learning environment that has the potential to enhance cognitive engagement.
- The study highlights the importance of using diverse analytical methods such as ENA and aRQA. Researchers may consider employing these methods to gain deeper insights into student engagement patterns and the efficacy of different instructional strategies.
- Institutions could consider offering professional development programs focused on the effective use of learning analytics and various analytical methods. Training educators on how to interpret and apply these analyses can enhance their instructional strategies and improve student engagement.

RESEARCH BACKGROUND

Theoretical background

Theory framework of the leaderboards

In this section, we outline the theoretical foundations related to leaderboards as a gamification motivational design element in online collaborative learning, explaining their potential roles in shaping students' learning.

Gamification, which involves applying game elements and mechanics to non-game contexts, has been widely used in OCL (Chen et al., 2020; Huang et al., 2019). Its core objective is to enhance students' motivation, engagement and behavioural performance by stimulating their interest in learning (Ding et al., 2017; Moon et al., 2024). Leaderboards, among the

most common and effective gamification elements, introduce fun and challenge into learning environments. They may not only arouse individuals' competitive consciousness but also enable members to understand the performance of their group through a dynamic feedback mechanism, thereby encouraging more active participation in collaborative learning (Chen & Chen, 2023; Lu et al., 2024). The introduction of leaderboards is well grounded in theory, mainly reflected in the following aspects.

Self-determination theory

This theory emphasizes the satisfaction of three basic psychological needs of individuals: autonomy, competence and relatedness (Ryan & Deci, 2000). Leaderboards visually display students' relative rankings, which may provide competence-related feedback and help reinforce their sense of competence. They may also facilitate social interaction, supporting the need for relatedness and fostering a sense of belonging that can motivate participation (Philpott & Son, 2022). However, excessive focus on external evaluation in leaderboards might erode autonomy, shifting motivation from intrinsic to extrinsic, which might potentially harm students' long-term engagement (Philpott & Son, 2022).

Social comparison theory

This theory suggests that individuals evaluate their abilities and self-worth through social comparison (Festinger, 1954). Leaderboards facilitate both upward and downward social comparisons through visible rankings. Upward comparisons may motivate improvement by inspiring users to emulate higher-ranked peers. However, perceived large gaps with top performers may cause frustration and demotivation. Downward comparisons, conversely, may boost self-confidence and satisfaction, strengthening engagement motivation (Bai et al., 2020).

In summary, leaderboards, as a gamification element, may contribute to enhancing learning by addressing learners' psychological needs and facilitating social comparison under appropriate conditions. However, the extent and nature of their educational effectiveness require further empirical examination. Accordingly, this study investigates the potential impact of leaderboards in OCL.

Cognitive engagement in OCL

Cognitive engagement, a crucial indicator of learning outcomes (Fredricks et al., 2004), refers to the degree of effort students exert in constructing knowledge in OCL (Kim et al., 2020) and reflects their internal psychological state (Kim et al., 2020). Studies have demonstrated that the quality of cognitive engagement can predict learning performance (Lei et al., 2018) and enhance group problem-solving capabilities (Xing et al., 2023). Analysing learners' discourse in OCL effectively explores their cognitive engagement (Chi & Wylie, 2014). The concept of *Cognitive Presence*, proposed by Garrison et al. (1999) within the theoretical framework of the Community of Inquiry, provides an analytical framework for examining cognitive engagement in OCL and has been widely applied in related research (Farrow et al., 2021). Cognitive presence is divided into four stages: *triggering event*, *exploration*, *integration* and *resolution*, with the first two stages representing lower-level cognitive engagement behaviours and the last two representing higher-level cognitive engagement behaviours (Farrow et al., 2021). Research indicates that higher levels of cognitive engagement behaviours significantly promote learning achievements (Liu et al., 2022).

However, a substantial body of research indicates that, without external interventions, student cognitive engagement in OCL tends to be low (Kristiana et al., 2023; Wu et al., 2025).

For instance, Wu et al. (2025) found that the cognitive engagement of cross-disciplinary groups significantly declines over time, increasingly relying on lower-order cognitive activities. This decline is detrimental to students' ability to achieve deep understanding and construct knowledge through communication and cooperation, and it does not contribute positively to learning performance or learning satisfaction (Chen et al., 2018). Therefore, maintaining and enhancing cognitive engagement is particularly crucial, and researchers have been exploring effective strategies and tools.

Overview of previous research

The importance of learning analytic feedback, leaderboards for students' cognitive engagement and learning performance

While Learning analytics feedback effectively enhances students' cognitive engagement and learning performance, its capacity to stimulate external competition and motivation remains limited—a gap that leaderboards may address. Learning analytics feedback employs an educational technology strategy to collect and analyse multidimensional learning process data, delivering visual diagnostic insights to learners (Echeverria et al., 2024; Ouyang & Zhang, 2024; Pozdniakov et al., 2021). In the OCL context, learning analytics feedback is typically presented in the form of group-based feedback, which means analysing the learning behaviours, cognitive processes and social interactions of the group, and then feeding the results back to group members in the form of charts or dashboards (Zheng et al., 2022, 2023). For instance, Zheng et al. (2022) utilized a deep neural network model to analyse online discussions and provide real-time feedback, significantly ameliorating learners' knowledge refinement, knowledge convergence, interactive relationships and group performance. Their subsequent work developed an automated method for analysing group learning engagement and providing feedback (Zheng et al., 2023) and validated its positive effects in enhancing learning engagement, group performance and collaborative knowledge building. However, existing research has shown that merely providing group-based learning analytics feedback is insufficient and may fail to motivate students to adopt the feedback and enhance the effectiveness of the feedback (Peng et al., 2022). Additional support for the group needs to be provided based on learning analytics feedback (Jivet et al., 2017; Strauß & Rummel, 2021).

Leaderboards, as a core gamification strategy, motivate students to set higher goals and enhance their drive for self-improvement by displaying achievements and progress, thereby stimulating comparisons among students (Balci et al., 2022). Sailer and Sailer (2021) demonstrated that when students engage in team-based quiz activities with visible leaderboards, this may support their performance during the learning process, which in turn has the potential to enhance their ability to apply knowledge in practice. Cigdem et al. (2024) found that integrating leaderboards into formative assessment procedures positively correlates with learner engagement and performance improvement. Additionally, when engaging with leaderboards, learners often become more aware of their own and their peers' performance gaps through social comparison and competition, which may help address some of the limitations of learning analytics feedback in conveying information about intergroup positions and gaps, thereby further promoting learning engagement (Bai et al., 2021). However, few empirical studies have examined the combined effects of learning analytic feedback and leaderboards on cognitive engagement and learning performance. Therefore, this study designed a learning analysis-based leaderboard feedback approach (LALF) to investigate its impact on students' cognitive engagement and learning performance.

Temporal analysis of the cognitive engagement

Cognitive engagement is a dynamic and complex developmental process in the OCL (Greene, 2015; Ouyang & Dai, 2022). Assessment methods must extend beyond traditional frequency statistics to incorporate temporal analysis for a deeper understanding of cognitive engagement (Zhou & Ye, 2024). The Epistemic network analysis (ENA) approach, proposed by Shaffer et al. (2009, 2016), is widely used for analysing network patterns. It reveals the interrelations between coding elements by quantifying the co-occurrence frequency of cognitive engagement behaviours, thereby supporting the construction of learners' cognitive patterns (Ba et al., 2023; Kong et al., 2025). In addition, auto-recurrence quantification analysis (aRQA) is a recursive quantitative analysis method designed for non-linear dynamic systems. It can identify the repetition of elements or patterns in a time series, revealing the regularity, predictability, stability, novelty and complexity of the data by quantifying these recurrence patterns (Dever et al., 2022; Li et al., 2022; Poquet et al., 2023). The aRQA quantifies the changing characteristics of recurrence patterns through multiple metrics, providing researchers with insights into the dynamic nature of the data. Some studies have proved that aRQA can effectively analyse the dynamic changes in learners' behaviours. Dever et al. (2022) employed aRQA to analyse learners' interactions with instructional materials, finding that predictable sequences were associated with longer dwell times and greater learning gains, indicating that participants with repetitive motion sequences achieved greater learning outcomes. They also used aRQA to assess the dynamic deployment of self-regulated learning (SRL) strategies among learners using intelligent tutoring systems, discovering that learners who received instructional agent prompts demonstrated higher learning effectiveness, lower repetition, and greater novelty in strategy use (Dever et al., 2023). Poquet et al. (2023) applied aRQA to analyse SRL behaviours in students enrolled in a blended undergraduate engineering course, capturing the complexity and regularity of student behaviour by quantifying changes in the entropy of their learning. In summary, ENA can analyse the pattern of student cognitive engagement, while aRQA allows further elaboration of the degree of change in the recurrence pattern. However, despite aRQA's potential, the application of aRQA to analyse cognitive engagement in OCL has been limited (Li, 2021), hindering the accurate capture and interpretation of dynamic changes in learners' cognitive engagement. Therefore, this study employed ENA and aRQA to identify and analyse the patterns and dynamics of learners' cognitive engagement in OCL, in addition to frequency counting. Based on the abovementioned research gap, this study proposed a learning analytics-based leaderboard feedback approach (LALF) and adopted the chi-squared test, ENA, and aRQA methods to analyse the effects of LALF on the distribution, patterns, and dynamics of cognitive engagement. In addition, this study investigates the impact of LALF on students' learning performance. We propose the following research questions:

Research Question 1. How does LALF affect students' cognitive engagement compared with traditional Learning Analytics Feedback (LAF)?

This is examined through three sub-questions:

RQ1.1. How does LALF affect the frequency distribution of behavioural codes related to student cognitive engagement in OCL compared to LAF?

RQ1.2. How does LALF influence the patterns of student cognitive engagement in OCL compared to LAF?

RQ1.3. How does LALF impact the dynamics of student cognitive engagement in OCL compared to LAF?

Research Question 2. How does LALF affect students' learning performance compared to LAF?

Drawing upon the theoretical background and current knowledge, the study proposes the following hypotheses:

H1.1. Compared with providing only learning analytics feedback, LALF enhances student interaction in OCL by fostering social comparison and competition, which in turn leads to high-level cognitive engagement behaviours.

H1.2. Compared with providing only learning analytics feedback, LALF enhances student interaction in OCL by fostering social comparison and competition, which in turn leads to stronger connections among high-level cognitive engagement behaviours.

H1.3. Compared with providing only learning analytics feedback, LALF enhances student interaction in OCL by fostering social comparison and competition, which in turn leads to higher stability of cognitive engagement.

H2. Compared with providing only learning analytics feedback, LALF enhances student interaction in OCL by fostering social comparison and competition, which in turn leads to higher learning performance.

METHODOLOGY

Ethics statement

This study adheres to the ethical standards of the Helsinki Declaration and received approval from the relevant institutional review board (Luo et al., 2023; Schoeps et al., 2018). During the study, participants had the right to freely join or withdraw at any time. Before the commencement of the study, all participants were fully informed of the purpose of the research, potential benefits and risks, and they signed informed consent forms. To ensure compliance with data usage, a data use agreement was signed between students and teachers, strictly protecting the privacy of students' personal information. Additionally, all data were anonymized, with sensitive identity information such as students' names and student IDs removed, to ensure the data were used solely for research purposes.

Participants and research context

The participants in this study consisted of 32 junior students majoring in software engineering at a university in China. All participants took part in the 'Object-Oriented Software Engineering' course during the spring semester of 2023. In the field of software engineering, teamwork is crucial; therefore, software engineering students must develop teamwork skills to better prepare for their future work environment (Lyu et al., 2023; Zhu et al., 2019). Therefore, this study adopted an online collaborative learning strategy. Although the sample size is relatively

small, through multiple collaborations, we have collected a substantial amount of detailed discourse data. The richness and granularity of these data provide a solid foundation for in-depth analysis (Ouyang et al., 2021). The tool supporting online collaborative learning activities in this study was QQ, as depicted in Figure 1 (Wu et al., 2025; Zhang et al., 2022). QQ enabled students to discuss collaboratively in real-time through its synchronous chat function. Additionally, it automatically recorded all dialogue, providing valuable data for analysing student cognitive engagement.

The learning analytics-based leaderboard feedback approach design

Building on the collaborative analytics conceptual model proposed by Martinez-Maldonado et al. (2021), we applied deep learning models, Gini coefficient calculations, and behavioural statistics to evaluate the group performance across cognitive, social, and behavioural dimensions, providing corresponding feedback (Li et al., 2021).

From a cognitive aspect, we utilized the BERT model (Luo et al., 2024) to automatically analyse the cognitive presence. In terms of behavioural metrics, we measured the total number of posts and the duration of discussions among each group (Zheng et al., 2023). At the social level, we assessed the balance of member participation in discussions to explore the quality of social interactions (Xing et al., 2023), employing the Gini coefficient to quantify this

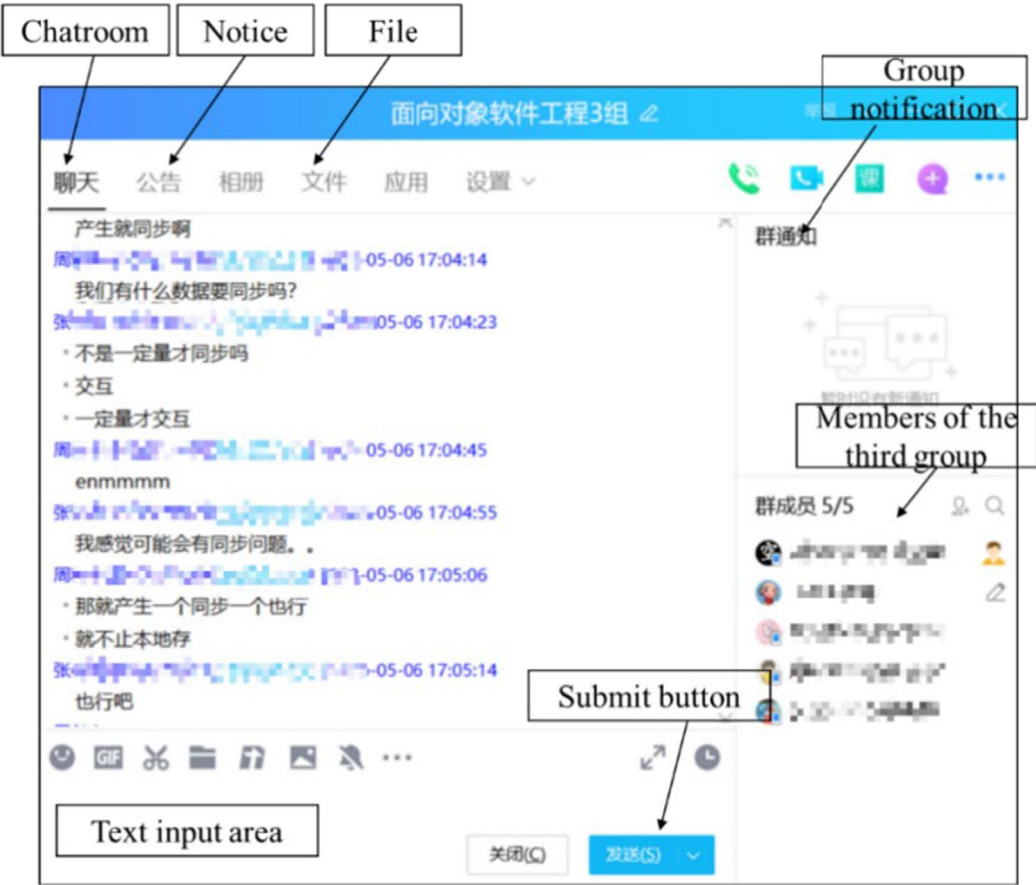


FIGURE 1 The QQ's interface.

balance. A Gini coefficient ranging from 0 to 0.2 indicates high group participation, reflecting a robust level of social engagement (Strauß & Rummel, 2021).

Based on these analysis results, we provided customized feedback for each group. Specifically, students in the CG received learning analytics feedback presented in the form of a dashboard, as illustrated in Figure 2.

In contrast to the CG, students in the EG received leaderboards in addition to the content of their learning analytics feedback. Displayed at the cohort level, the leaderboard ranks entire learner groups on their collective performance, thereby stimulating individuals to engage in group-level competitive behaviours, such as improving the group's cognitive performance. The design of the leaderboards primarily drew upon the research of Philpott and Son (2022). The scores and rankings of the leaderboard are based on the results of the learning analytics and are further refined according to specific rules, which are detailed in Table S1. Two distinct leaderboards were established: a weekly leaderboard and an overall leaderboard, as illustrated in Figure 3(a,b), respectively. The weekly leaderboard displays the scores of each group for each week, while the overall leaderboard shows the cumulative scores of all groups. It is important to note that the leaderboard is accessible only to students in the EG, and students in the CG were not permitted to view their scores to ensure the validity of the experiment.

Experimental procedure

To explore the impact of LALF on student cognitive engagement and learning performance, this study adopted a quasi-experimental design. In the field of educational research, quasi-experimental design is widely used to assess the effectiveness of teaching methods and interventions (Johannes & Haase, 2024; Peng et al., 2022). This design is particularly suitable for implementation in natural teaching environments, as it can provide valuable causal inferences without fully controlling all variables.



FIGURE 2 The screenshot of the CG's learning analytics feedback. Illustrates the feedback received by Group 1 before their discussion in Week 8.

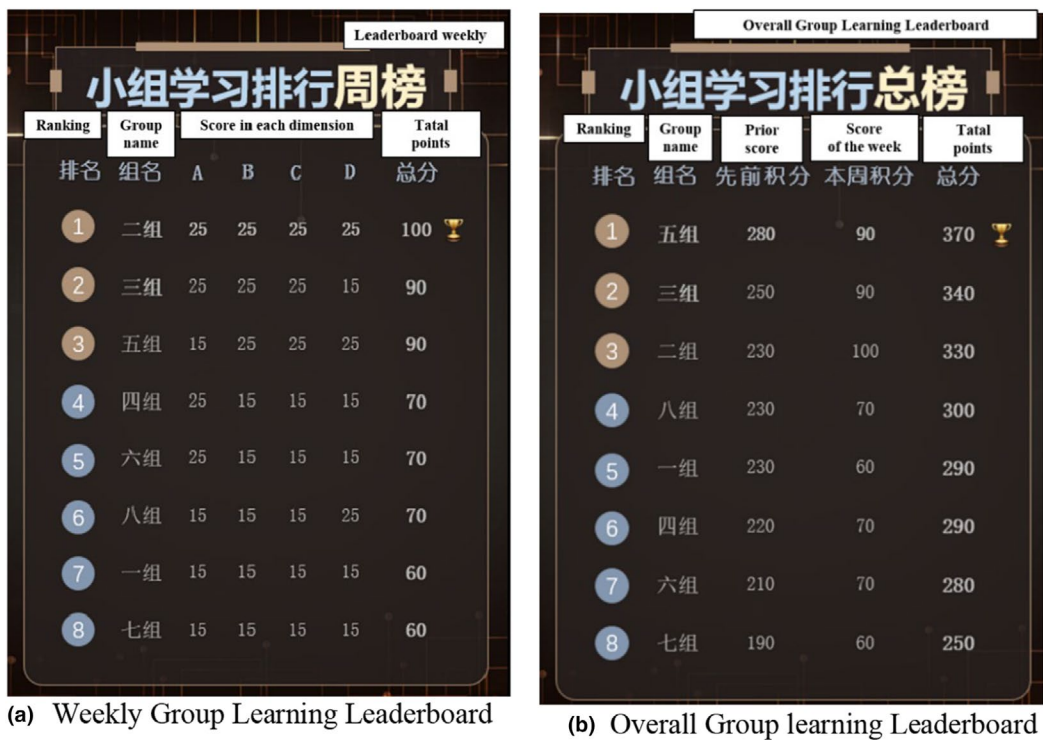


FIGURE 3 The screenshot of the EG's leaderboards.

The participants in this study were 32 engineering students who independently formed eight collaborative learning groups, with four students in each group. This grouping method is based on the natural grouping patterns in real teaching contexts, where students can freely form groups according to social relationships or interest preferences (Wu et al., 2025). While this natural grouping approach aligns with actual teaching scenarios, it also introduces a key characteristic of quasi-experimental design: the inability to fully randomly assign participants (Cook et al., 1979). This uncontrollable aspect of the grouping method is an important distinction between quasi-experimental design and randomized controlled trials (RCTs). It limits our ability to randomly assign participants, thereby affecting the internal validity of the study.

After the natural grouping was completed, we randomly assigned the eight groups into experimental and control groups, with four groups in each, and a total of 16 participants in each group. Specifically, we used a random number generator to randomly number the eight groups and then assigned them to the experimental and control groups based on the numbers. This random assignment process helps to some extent to increase the similarity between the two groups in terms of initial conditions, although there may be some uncontrollable factors in the initial natural grouping.

Figure 4 illustrates the experimental procedure. Before the experiment, all students were introduced to the primary task of the course, which was to complete the design and implementation of a learning analytics system. In addition, we distributed a 5-point Likert scale to assess students' perceptions of their own groups' collective efficacy (Wang & Lin, 2007) for completing the task (ranging from 1=strongly disagree to 5=strongly agree). A *t*-test revealed no significant differences between the two groups ($p=0.301>0.05$). This result provides some evidence for the comparability of the two groups' baseline levels before the intervention (Peng et al., 2022). During the experiment, neither the EG nor the CG received any intervention for the first 1–5 weeks. Subsequently, during Weeks 6–10, students in the

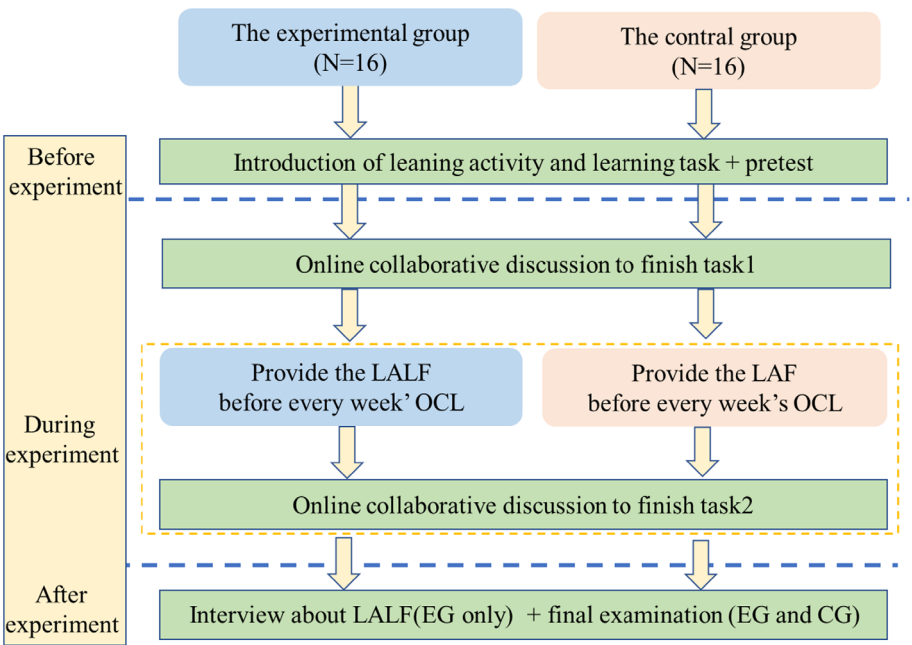


FIGURE 4 The experimental procedure.

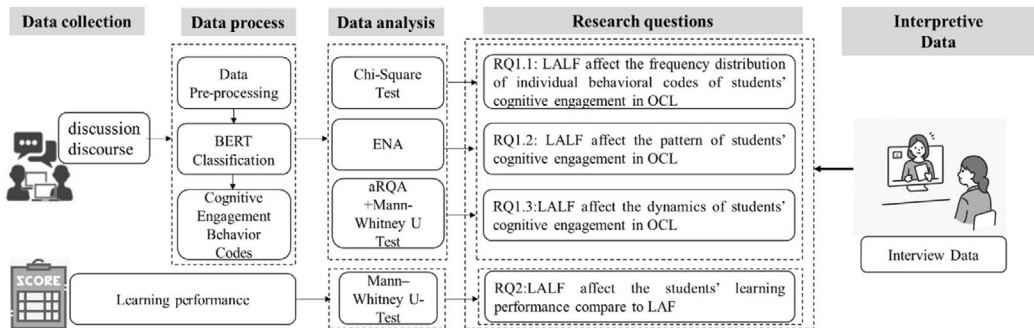


FIGURE 5 The design of data collection and analysis.

CG only received learning analytics feedback, while students in the EG received LALF before their weekly discussion activities. After the experiment, we collected the learning performance data of the participants. In addition, to further analyse how LALF affects students' cognitive engagement and learning performance, we conducted interviews with the students in the experimental group.

Data collection and analysis

As shown in Figure 5, this study primarily collected three types of data: discourse data, learning performance and interview data. Firstly, we primarily focused on discourse data generated by students during online collaborative discussions in Weeks 6–10. The data processing involved three critical stages: Initially, text standardization including removal of non-textual content (images, emojis) and normalization of internet slang expressions. Next,

the application of our team's developed classification model (Luo et al., 2024) for automated discourse coding. This model was trained on historical data from the same course and demonstrated strong inter-rater reliability through random sampling validation. Lastly, these discourse data were classified into five cognitive engagement behaviours (see Table 1), resulting in a total of 6057 cognitive engagement behaviour codes.

Secondly, we collected the participants' learning performance data, which was measured by the final examination scores. The purpose was to assess students' mastery of the course content through a standardized test. One control group student missed the final examination, resulting in learning performance data from 31 participants.

Finally, we conducted interviews with students from the EG to explore how LALF affects students' cognitive engagement and learning performance. Before the formal interviews, we introduced the LALF to the corresponding students to ensure a full understanding of the relevant concepts. The main questions of the interview outline can be found in Appendix A.

This study employs individual students as the unit of analysis. Although the rankings present comparative data at the group level (eg, overall group participation and cognitive engagement distribution), the underlying influence mechanism operates through social comparison, competition and goal-setting on individual students' motivation and behaviours (Festinger, 1954). This pathway of influence from group information to individuals has been supported by empirical research (Ollesch et al., 2021; Peng et al., 2022). All statistical tests in this study are conducted with the individual as the unit of analysis.

To address Research Question 1, this study sequentially employed three analytical methods (chi-squared test, ENA, and aRQA) to conduct comparative analyses of the frequency, patterns, and temporal dynamics of cognitive engagement behaviours between students in the EG and CG. The uniformly processed cognitive engagement behaviour coding data underwent method-specific transformations to meet each analytical method's requirements.

For RQ1.1, we used SPSS 26.0 to conduct a chi-square test to explore differences in the frequency distribution of cognitive engagement behaviours, with input data consisting of a cross-table including the frequency distributions of five cognitive engagement behaviour codes between students in the EG and CG.

For RQ1.2, we employed an online tool for ENA (<https://www.epistemicnetwork.org/>) to analyse structured data sets containing discourse content, timestamps, dates and corresponding codes from both groups (Shaffer et al., 2016). Detailed format specifications are provided in Table S2. The ENA analysis configured 'Type (EG/CG)-Group Id—Student Id' as the unit of analysis, 'Date' as the conversation and the five cognitive engagement behaviours as coding elements, ultimately generating comparative visualizations of the distinct cognitive engagement patterns between the students in the EG and CG.

TABLE 1 Examples of cognitive engagement behaviours.

Category	Examples
Triggering event	'Can it be used to analyse the student learning process?'
Exploration	'I think we should see if there are any improved algorithms regarding the prediction of academic performance'
Integration	'This is what we have come up with so far. (1) Student per grade data file; (2) student paper question score data file; (3) student unit work completion; (4) student video learning; (5) student posting data file'
Resolution	'Okay, following our idea, we're going to add the academic performance prediction function to our system'
Other	I am not in school now

For RQ1.3, we performed aRQA using the 'casnet' package in R (Poquet et al., 2023), with input data consisting of time-sequenced cognitive engagement behaviour codes from the students in the EG and CG. Readers can find more details about the mathematical and computational foundations of the method in the research by Marwan and Webber (2015). The aRQA provides multiple indicators to quantify recurring patterns, with various terms used to describe these measurements, including but not limited to regularity, predictability, stability, novelty and complexity (Dever et al., 2022; Li et al., 2022; Poquet et al., 2023). In this paper, we focused on the term *stability* to compare the degree of dynamic change in cognitive engagement between the EG and CG. Following aRQA of these behavioural code sequences from students in the EG and CG, we extracted four metrics: Recurrence Rate, Entropy, Laminarity, and Determinism to explore the stability of learners' cognitive engagement (Dever et al., 2022; Dever et al., 2023; Li et al., 2022; Poquet et al., 2023). Recurrence rate refers to the frequency at which specific behaviours repeat in a time series. A higher recurrence rate indicates that learners are more likely to reuse certain patterns. Entropy is calculated by quantifying the distribution of recurrence patterns. Specifically, entropy measures the uniformity of the distribution of recurrence pattern lengths. If the distribution of recurrence pattern lengths in the time series is very uniform, the entropy value will be high, indicating that the system is in a high-entropy state, meaning that the behaviour is more uncertain or random. Conversely, if the distribution of recurrence pattern lengths is more concentrated, the entropy value will be lower, indicating that the system is in a low-entropy state, meaning that the behaviour is more predictable. Laminarity reveals the extent to which patterns of the same cognitive engagement behaviour repeat. Determinism indicates the frequency of how often a sequence of behaviours is repeated in a fixed order. Higher values of determinism indicate a more structured and predictable pattern of behaviour, reflecting the orderly repetition of behaviour. Following the derivation of these metrics through aRQA analysis, and given the non-normal distribution of these stability metrics, Mann–Whitney U tests were employed to analyse, at the individual student level, how the stability of participants' cognitive engagement behaviours varied across feedback conditions.

Additionally, aRQA constructs recurrence plots by mapping time series data into a two-dimensional matrix, revealing the recurring dynamic structures within the data. A recurrence plot is a visualization tool that illustrates the internal dynamics of a system by showcasing the repetition of patterns in time-series data. As shown in Figure 6, the recurrence plot of aRQA features the same time series along both the X-axis and the Y-axis. In this plot, black squares, also known as recurrence points, indicate behaviours or states that recur at specific time points in the sequence. These black squares are arranged along the diagonal, visually demonstrating the repetitive patterns of behaviours or states in the time series. Furthermore, the recurrence of two consecutive behaviours is represented by a diagonal line of length 2 in the plot (see Figure 6A); similarly, the recurrence of three consecutive states forms a diagonal line consisting of three points (see Figure 6B). The distribution of the lengths of these diagonal lines reflects the predictability or complexity of the recurrence patterns within the system, with longer black diagonal lines indicating a more repetitive and predictable system. Therefore, in addition to analysing quantitative indicators, we also examined the recurrence plots to gain further insights into the dynamic changes in students' cognitive engagement.

To address RQ2 regarding the impact of LALF on learning performance, we conducted Mann–Whitney U tests using SPSS 26.0 to compare achievement outcomes between the EG and CG at the individual student level, as the students' performance data violated the normality assumption.

In addition to supporting triangulation and interpretation of the quantitative results, we followed Braun and Clarke's (2006) methodology and applied thematic analysis to conduct an in-depth examination of the interview data.

RESULTS

The effect of LALF on the frequency distribution of cognitive engagement

Table 2 presents the frequency and percentage of cognitive engagement among students in the EG and CG. The results showed that the EG exhibited a higher proportion of behavioural coding in *Exploration*, *Integration* and *Resolution* compared to the CG. Conversely, the EG

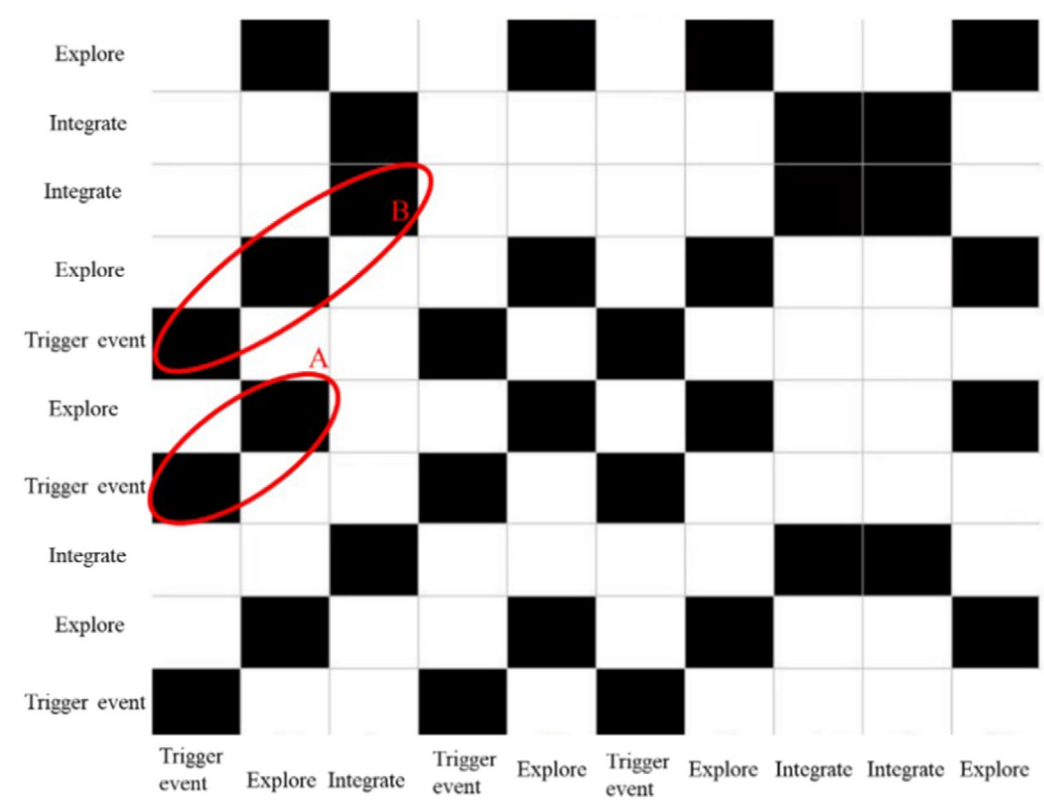


FIGURE 6 Recurrence plots.

TABLE 2 Comparison of cognitive engagement codes distribution between EG and CG with overall chi-square test results.

Codes	EG		CG		χ^2	p
	(F)	(%)	(F)	(%)		
Triggering event	787	18.86	396	21.02	65.45	0.000
Exploration	2583	61.90	1125	59.71		
Integration	583	13.97	200	10.62		
Resolution	87	2.08	27	1.43		
Other	133	3.19	136	7.22		

Abbreviations: %, proportional distribution; F, frequency [absolute count].

demonstrated a lower proportion of coding in *Triggering events* and *Other* behaviours than the CG. Overall, both groups displayed a positive exploratory stance during the discussion process, which is essential for completing collaborative learning tasks. The overall chi-squared analysis revealed significant differences between the experimental and control groups across categories of cognitive engagement behaviours ($X^2=65.45$, $df=4$, $p<0.001$). In addition to reporting the significance level of the chi-squared test, we also calculated the effect size using Cramer's V ($V=0.104$), which indicates a small effect according to Cohen's guidelines (Cohen, 1988; Sun et al., 2010). It indicates relatively minor differences in the proportional distribution of the five cognitive engagement behaviours between the EG and CG. Notably, as shown in Table 2, the absolute frequencies (F values) of all cognitive engagement behaviour codes in the EG were nearly double those of the CG. This difference may suggest that students in the EG demonstrated higher levels of cognitive engagement compared with those in the CG.

The effect of LALF on the pattern of cognitive engagement

This study employed the ENA to explore the differences in cognitive engagement patterns between the EG and CG. Figure 7(a,b) illustrates the cognitive networks of learners in the EG and CG, respectively. The results indicate that both groups exhibit strong connections between '*Trigger event–Exploration*' and '*Exploration–Integration*'. Figure 7(c) presents a comparison of the average networks for learners in the EG and CG. We can see that the X-axis accounts for 49.3% of the data variance, whereas the Y-axis accounts for 19.4%. Learners in the EG are represented by red dots in the ENA space, while learners in the CG are represented by blue dots. The red and blue squares represent the centroids (ie, the

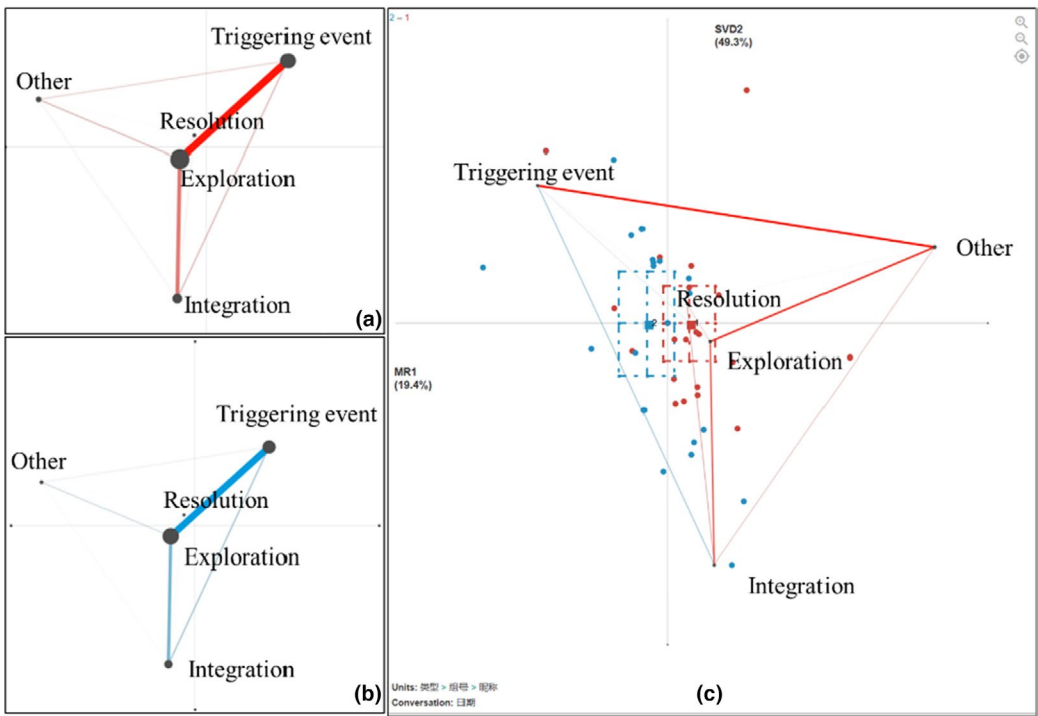


FIGURE 7 Comparison of cognitive engagement networks of EG (red) and CG (blue).

average position of the projection points) of the learners in the EG and CG, respectively. The centroid of the CG learners is located on the left side of the coordinate axis, whereas the centroid of the EG learners is on the right side.

The connection weights in the figure represent the strength of the relationships between the cognitive engagement behaviour coding of learners in the EG and CG, with line colours indicating learners with close connections. The thickness of the lines correlates with the strength of the connections between the corresponding coding elements. The study results demonstrate that, compared to CG learners, EG learners exhibit stronger connections among cognitive engagement behaviours such as ‘*Trigger event–Other*’, ‘*Exploration–Integration*’, ‘*Exploration–Other*’ and ‘*Integration–Resolution*’. In contrast, CG learners show a stronger connection in ‘*Trigger Event–Integration*’. The Mann–Whitney *U* test was used to statistically verify the differences between the EG and CG. The test results indicate that at a significance level of $\alpha=0.05$, the distribution difference on the X-axis is statistically significant ($U=330.50$, $p=0.02$, $r=0.43$), whereas the distribution difference on the Y-axis is not significant ($U=213.5$, $p=0.68$, $r=0.08$).

To explicitly illustrate differences in cognitive engagement patterns between the EG and CG, we conducted a micro-level discourse analysis (Zhang et al., 2022). Detailed information for each group was presented in [Appendix B](#).

The effect of LALF on the dynamics of cognitive engagement

The study employed aRQA and the Mann–Whitney *U* test to compare the differences in the dynamic changes in cognitive engagement behaviours of students in the EG and CG, with the detailed results presented in [Table 3](#). The findings revealed that students in the EG exhibited significantly lower entropy values compared to the CG ($p<0.05$). In addition, the EG demonstrated marginally better performance on recurrence rate, laminarity and determinism metrics, though these differences did not reach statistical significance. Notably, the lower entropy values coupled with elevated recurrence rate, laminarity, and determinism suggest more stable cognitive engagement behaviours among learners. These results suggest that students who received the LALF exhibited more stable patterns of cognitive engagement behaviours than the students who received the traditional learning analytics feedback. Furthermore, their lower standard deviations across all metrics compared to the control group indicate that these learners tended to maintain more consistent levels of engagement throughout the collaborative learning process.

The statistical inference (Mann–Whitney *U* test) is rigorously derived from formal analyses conducted at the individual level, whereas randomly selected individual cases were utilized to illustrate the micro-level dynamic changes of cognitive engagement behaviours. We randomly selected one student each from the EG and CG, then generated their recurrence plots ([Figure 8](#)) to elucidate the dynamics of cognitive engagement in the two groups. From the results, the recurrence plot of the EG student shows a higher

TABLE 3 The metrics of the aRQA analysis.

aRQA metrics	CG <i>M</i> (SD)	EG <i>M</i> (SD)	<i>U</i>	<i>Z</i>	<i>p</i>
Recurrence rate	0.44 (0.10)	0.45 (0.06)	108	−1.01	0.313
Entropy	0.31 (0.11)	0.26 (0.06)	79	−2.05	0.04*
Laminarity	0.78 (0.16)	0.83 (0.06)	122	−0.50	0.61
Determinism	0.71 (0.14)	0.74 (0.07)	135	−0.036	0.971

*indicates statistical significance at $p\leq0.05$.

number of black squares and a distribution along the diagonal line, indicating a propensity for the student to repeatedly engage in a certain cognitive engagement behaviour or behaviour pattern. In contrast, the recurrence plot of the control group exhibits a more dispersed distribution of black squares, which may suggest less stable behaviour patterns among CG students.

The effect of LALF on students' learning performance

To address RQ2, we employed the Mann–Whitney *U* test to compare the differences in students' learning performance between the EG and CG, with detailed results presented in Table 4. Table 4 shows that EG students performed significantly better than CG students ($U=62.50$, $Z=-2.94$, $p<0.05$). This suggests that students who received LALF tended to achieve higher learning performance than those in the CG.

DISCUSSION

Addressing the research questions

Firstly, the students in the experimental group demonstrated higher levels of cognitive engagement than those in the control group. Specifically, compared with the CG who

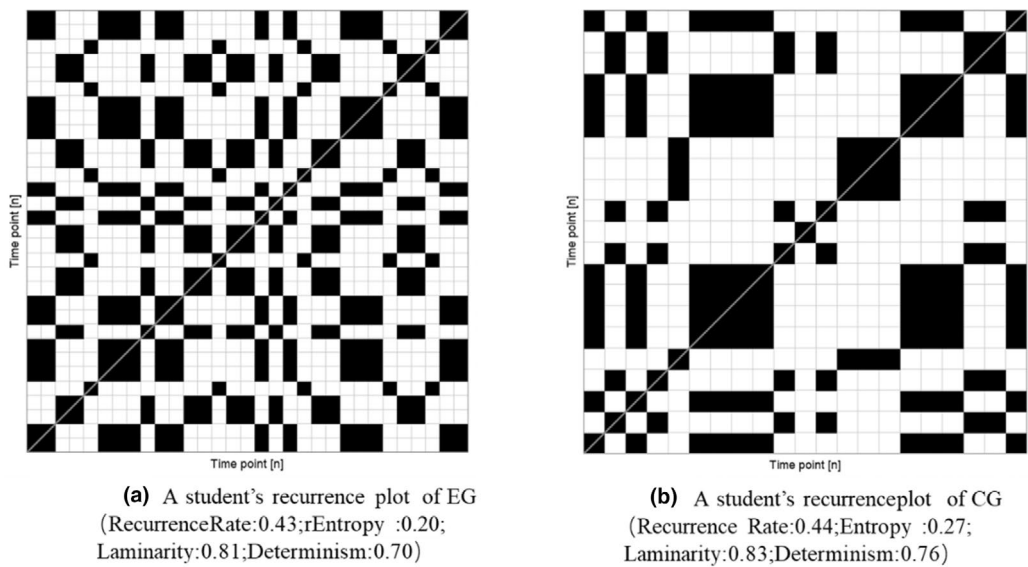


FIGURE 8 Comparison of recurrence plots of EG and CG.

TABLE 4 Students' learning performance of EG and CG.

Variable	Group	N	Mean	SD	U	Z	p
Performance	EG	16	84.81	5.91	62.50	-2.275	0.023*
	CG	15	82.25	2.73			

*indicates statistical significance at $p\leq0.05$.

only received learning analytics feedback, the EG students were more frequently involved in high-level cognitive activities such as *Exploration*, *Integration* and *Resolution*. This finding aligns with the research of Cigdem et al. (2024), who demonstrated that leaderboards can effectively promote learning engagement. Furthermore, this discovery extends and deepens existing research by showing that the students who engage with learning analytics feedback combined with leaderboards may engage more deeply in cognitive activities. This may be because leaderboards provide additional incentives and comparison benchmarks, stimulating competition among students and thereby promoting deeper learning and understanding (Balci et al., 2022). For instance, one interviewee stated that he frequently checked the leaderboard and increased his speaking frequency to improve his group's ranking when they found his group lagging. Another interviewee stated 'By examining the exploration graphs, trigger events, and other metrics showing our group's weekly progress, I would check the leaderboard, see we were second again, compare with the top group to identify gaps and adjust my cognitive strategy accordingly'. This phenomenon can also be explained by the Goal-Setting Theory, which posits that clear, specific and appropriately challenging goals can effectively motivate individuals (Locke & Latham, 2002). In this study, learners appeared to set clear and challenging goals for themselves by viewing the leaderboard and related metrics, namely, to improve their ranking or close the gap with the leading group. The establishment of such goals has stimulated their learning motivation and prompted them to actively adjust their learning strategies. Nevertheless, previous studies have noted that excessive or unfavourable comparisons may sometimes lead to frustration or reduced motivation, suggesting that leaderboard design should balance competitiveness with supportive feedback (Bai et al., 2020; Landers et al., 2017).

Secondly, the EG students appeared to demonstrate stronger connections among high-level cognitive engagement behaviours than the CG. Through ENA, we found that the EG had stronger connections between high-level cognitive engagement behaviours such as '*Exploration-Integration*' and '*Integration-Resolve*'. This finding is consistent with the higher performance of the EG in the proportion of integration and resolution coding in the frequency analysis, suggesting that learners who received leaderboard-supported feedback demonstrated stronger connections between high-level cognitive engagement behaviours. Previous research has emphasized the importance of connections between high-level cognitive engagement behaviours (Wu et al., 2025). Wu et al. (2025) found that students in interdisciplinary groups had more frequent connections between low-order cognitive engagement behaviours and relatively fewer connections between high-order cognitive engagement behaviours, indicating that interdisciplinary groups faced challenges in promoting high-level cognitive engagement behaviours. In contrast, the strong connections between high-level cognitive engagement behaviours among the EG students in this study may reflect that leaderboard feedback based on learning analytics, as an incentive mechanism, was associated with students' deep participation in cognitive activities and more complex patterns of thinking. This suggests that the approach we proposed not only increased students' participation in high-level cognitive activities but also strengthened the interconnections between these activities, which may indicate deeper learning and understanding.

Thirdly, the students in the EG maintained more stable cognitive engagement patterns throughout the OCL process compared with those in the control group. Although previous research has indicated that the level of students' cognitive engagement tends to decrease over time during online discussions (Liu et al., 2021; Wu et al., 2025), this study found that learners in the EG were able to sustain relatively stable cognitive engagement throughout the discussion process. The results highlight the potential of leaderboard feedback based on learning analytics as a teaching tool, which helps maintain the stability of learners' cognitive engagement patterns in collaborative learning environments by providing continuous

incentives and feedback. This finding suggests that leaderboard feedback based on learning analytics may help support the continuity of learners' cognitive engagement by providing ongoing incentives and feedback. Stability enables learners to consistently apply effective learning strategies, which have been associated with deep learning and understanding (Dever et al., 2022). Such stability may partly explain the higher levels of cognitive engagement observed in the EG. Therefore, these findings highlight the potential value of integrating learning analytics and leaderboard feedback in supporting more stable cognitive engagement patterns, which could inform the design of future collaborative learning environments.

Fourth, the students in the EG tended to perform better than those in the CG in terms of learning performance. This aligns with Sailer and Sailer's (2021) experimental demonstration that leaderboard-enhanced gamification in flipped classrooms improved education science students' outcomes. Specifically, leaderboards influenced application-oriented knowledge through the mediating variable of performance in the learning process. Similarly, Cigdem et al. (2024) reported that integrating leaderboards as a gamification element into formative assessment procedures enhanced student engagement and learning performance. Previous research has explored the relationship between cognitive engagement and learning outcomes, revealing that high-level cognitive activities such as integration and resolution can promote learning outcomes (Liu et al., 2022). In this study, one student from the EG mentioned: "To improve our grades, we need to be more proactive." This statement indicates that students supported by LALF demonstrated higher levels of engagement, which may account for their superior performance compared to those in the CG. Our study provides new insights into this relationship by showing that students who received learning analytics-based leaderboard feedback tended to engage more deeply in cognitive activities, providing empirical evidence for its potential as a supportive teaching strategy.

Implications

From a practical perspective, the findings of this study offer meaningful insights for teaching and research practices. The LALF approach proposed in this study provides new ideas for educational practice. Beyond our technical implementation, practitioners could adapt this feedback approach through multiple applications. For instance, teachers might link leaderboards to learning progress by tracking task completion time or quality and converting these metrics into leaderboard scores. Early, high-quality task completion could earn higher rankings, motivating accelerated progress and improved outcomes. Moreover, teachers could use data from learning management systems (LMS), such as login frequency, study time, and discussion participation, to inform leaderboard metrics that may help encourage better learning habits. Consistent engagement could earn higher rankings, reinforcing positive u. For innovation-focused courses, leaderboards might assess creative projects, original solutions, and knowledge exploration to motivate inventive thinking. Such feedback stimulates creativity and cultivates innovative thinking. These examples demonstrate how practitioners can adapt LALF to course-specific objectives.

In terms of research practice, when conducting a time-series analysis of learners' cognitive process data, it is essential to consider not only the cognitive patterns but also their stability. By analysing the stability of learners' behavioural patterns, educators can identify cognitive bottlenecks or motivational fluctuations that students may encounter at specific learning stages. For instance, if the analysis reveals that a group initially exhibits a high degree of '*Trigger event–Exploration–Integration–Resolution*' behaviour patterns, but these patterns become unstable over time, it may indicate that students are facing difficulties or losing interest. Based on these data, teachers can intervene on time, provide personalized guidance, stimulate students' intrinsic motivation or adjust teaching strategies to

better meet the learning needs. Additionally, the analysis of behavioural pattern stability offers a new perspective for educational research, encouraging researchers to explore how different teaching strategies affect learners' behaviour and how to optimize learning outcomes by adjusting teaching methods. Therefore, the application of the aRQA method can help shift educational practice from a one-size-fits-all teaching model to more flexible and personalized teaching approaches, ultimately improving teaching quality and learning outcomes.

From a theoretical perspective, this study introduced leaderboard feedback into the OCL environment based on Social Comparison Theory, Goal-Setting Theory, and Self-Determination Theory. The results indicate that this intervention may have contributed to higher levels of cognitive engagement and better learning performance. Thus, this study empirically validated the effectiveness of these theories in practice and extended their applicability to digital collaborative learning environments, thereby further deepening and expanding these theories. Additionally, this study deepened the design theory of gamified learning. By combining learning analytics technology with traditional leaderboards, we expanded the educational function of gamification elements, transforming them from simple competitive motivational tools to multidimensional feedback systems that support in-depth cognitive engagement. This transformation enables gamified design to meet both surface-level behavioural motivation and deep-level cognitive development needs, providing a reference for future educational technology design. Of course, the specific mechanisms of these theoretical applications still need to be further explored in subsequent research.

Limitations and future directions

Despite the meaningful findings, the study has five limitations that suggest directions for future research. First, the sample size of only 32 junior software engineering students is relatively small, which may limit the generalizability of the results. Future research could validate the effects of LALF with a larger sample to further confirm its effectiveness. Second, as this study focused exclusively on software engineering students, future research should explore LALF's effects across disciplines and course settings. Third, due to privacy protection regulations, we were unable to collect students' GPA data as baseline measurements and instead employed the perception of their own group's task completion capability as a proxy measure. While this approach is theoretically grounded and academically recognized, it cannot completely rule out potential differences in academic levels between groups. Future studies should enhance the reliability of baseline data by collecting GPA or standardized pretest scores while adhering to ethical guidelines. Fourth, this study mainly relied on students' discussion data to assess cognitive engagement, which may have certain limitations. Future research could incorporate additional measurement tools, such as behavioural observation, EEG analysis and eye-tracking analysis, to provide a more comprehensive evaluation of the effects of LALF. Finally, although the quasi-experimental design employed in this study provides a feasible pathway for causal inference in real educational settings, the internal validity of the study conclusions still needs to be treated with caution due to the limitations of non-random assignment and the potential influence of confounding variables. Future research may further validate and extend the findings of this study by adopting mixed method designs, introducing stricter statistical controls (such as difference-in-differences or instrumental variables), or conducting randomized controlled trials when conditions permit.

CONCLUSION

This study provides important empirical evidence for understanding the impact of the learning analytics-based leaderboard feedback approach (LALF) on online collaborative learning effectiveness. By combining data-driven insights from learning analytics with the motivational aspects of leaderboards, LALF represents a novel instructional approach that supports cognitive engagement. Utilizing a quasi-experimental design, this study allocated 32 junior software engineering students into either an EG or a CG, deploying distinct feedback approaches throughout a software engineering curriculum. Our data suggest that students who engaged with LALF demonstrated higher levels of cognitive engagement, which appears to be associated with social comparison and competitive behaviours prompted by using the leaderboard.

Importantly, this study extends previous research by systematically showing how integrating learning analytics with gamified leaderboard feedback can promote sustained high-level cognitive engagement, providing empirical evidence for its potential as a teaching strategy, which has not been demonstrated in prior studies.

Although this study has certain limitations, such as a small sample size and a focus on students in software engineering, which may restrict the generalizability of the findings, these limitations do not diminish the value of the results. Firstly, the study employed multiple analytical methods, including chi-squared tests, ENA and aRQA, to enhance the internal validity of the results. This multi-method approach provides supportive evidence for the effectiveness of the LALF approach. Secondly, although the study primarily relied on discussion data to assess cognitive engagement, which may have certain limitations, the combination of various analytical methods still enabled us to capture the key characteristics and dynamic changes of cognitive engagement. The contribution of this study lies in proposing the learning analytics-based leaderboard feedback approach, which provides empirical evidence for designing effective online learning tools and offers methodological references for future research through the use of multiple analytical methods. These findings offer new empirical support for enhancing cognitive engagement in OCL and lay an important foundation for future research.

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CONFLICT OF INTEREST STATEMENT

None.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The Ethic Committee of Central China Normal University has approved this study (CCNU-IRB-202302032b).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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APPENDIX A

THE QUESTIONS OF THE INTERVIEW

The main questions of the interview outline included: (1) Are you willing to view and able to understand this feedback information? (2) What potential impacts have these feedbacks had on you or your group? (3) What factors do you think will affect your use of this feedback? (4) What potential impacts has the leaderboard had on you or your group? and (5) How did you feel when the leaderboard was added?

APPENDIX B

THE MICRO-LEVEL DISCOURSE ANALYSIS OF STUDENTS IN EG AND CG

Table B1 excerpts a typical experimental group conversation. In the conversation, Student E1 effectively guided the discussion by posing a question. Student E2 initially provided a brief social response and then systematically elaborated on the key tasks of requirements analysis, including core aspects such as clarifying requirements and feasibility assessment. Building on this, Student E3 added important dimensions, such as maintainability and testability. Student E4 then integrated the viewpoints of the previous members and systematically organized the discussion content. Finally, the discussion concluded with a clear consensus statement by Student E1, making the entire discussion process natural and efficient. This micro-level example reveals the strong connections between EG students' cognitive engagement behaviors of 'Trigger event- Other', 'Exploration-Integration', 'Exploration-Other', and 'Integration- Resolution'.

TABLE B1 Example of discourse record from the experimental group discussion.

Turn	Discourse record
1	Student E1: E2, can you share your thoughts? (Code: <i>Triggering event</i>)
2	Student E2: I am coming (Code: <i>Other</i>)
3	Student E2: In object-oriented software engineering, requirements analysis is one of the important stages in the software development process. It involves the early phase of software development and aims to clarify the requirements of the software system to facilitate subsequent design, coding and testing. The main tasks in the requirements analysis stage include: (1) Determining requirements: Clarifying the problems that the software system needs to solve, and sorting out user requirements and business requirements. (2) Analysing requirements: Conducting in-depth analysis of requirements, including assessing feasibility, priority, stability, etc., and abstracting and generalizing requirements. (3) Confirming requirements: Communicating with users and relevant stakeholders to ensure the accuracy and completeness of requirements. (4) Documenting requirements: Writing requirements into documents, including functional descriptions, business processes, use case analysis, user requirements, etc. (Code: <i>Exploration</i>)
4	Student E3: I agree with the points mentioned above. In addition, I think requirements analysis should also consider the following aspects: (1) Maintainability: Requirements should consider the maintainability of the system to facilitate effective maintenance and repair after software development. (2) Testability: Requirements should be testable and verifiable to ensure software quality and reliability during development. (3) Establishing a mechanism for requirement changes: During requirements analysis, it is necessary to establish a mechanism and process for requirement changes to manage them effectively during development. (4) Identifying user roles and scenarios: Requirements analysis should clarify user roles and scenarios to meet different user needs effectively. (5) Interaction design: Requirements analysis should consider the interaction design of the system to provide a good user experience and usability. (Code: <i>Exploration</i>)
5	Student E4: Let me integrate everyone's viewpoints. The requirements analysis stage should not only focus on clarifying and documenting basic requirements but also consider factors such as system maintainability, testability, requirement change mechanisms, user roles and scenarios and interaction design. This ensures that the software system can be efficiently and high-quality developed and meet long-term user and business needs. (Code: <i>Integration</i>)
6	E1: Let us go with this, then. We will conduct the requirements analysis from these dimensions. (Code: <i>Resolution</i>)

TABLE B2 Example of discourse record from the control group discussion.

Turn	Discourse record
1	Student C1: C2, can you share your thoughts? (Code: <i>Triggering Event</i>)
2	Student C2: For the student's end, I can only think of the teacher sending emails for now (Code: <i>Exploration</i>).
3	Student C3: That works; learners can be alerted that way (Code: <i>Integration</i>)
4	Student C3: What do you think, @C4? (Code: <i>Triggering Event</i>)
5	Student C4: I agree with the previous idea. Maybe we could also create an interface to connect with the SPOC platform? (Code: <i>Integration</i>)
6	Student C4: What do you guys think? (Code: <i>Triggering Event</i>)
7	Student C1: I am not quite sure. Let us hear everyone's opinions. Is that okay? (Code: <i>Triggering Event</i>)

The example in [Table B2](#) illustrates a typical discussion segment from the control group, focusing on system function design. Specifically, Student C1 initiated the discussion by posing a question. Student C2 then provided a relatively simple suggestion, demonstrating limited depth of exploration. Student C3 attempted integration but abruptly shifted to ask others, disrupting the discussion flow. Student C4 introduced a new idea but concluded with another question. Ultimately, Student C1 expressed uncertainty, and the discussion failed to reach a clear conclusion. This reveals lower motivation in some control group students, requiring peer reminders that disrupt content integration and consensus-building. This micro-level example reveals the strong connections between CG students' cognitive engagement behaviours of '*Integration-Triggering event*'.